

MTH 346/646 - Class 21 - 11/6

Fun fact: Today is 11/6, which is equal to $1 + 1/2 + 1/3$.

Last time: We did examples of $y^2 = x^3 + 13x$, talked about why it's hard to find the rank of $y^2 = x^3 + 17x$, talked about solving $1^3 + \dots + n^3 = m^6$, and talked through using Magma to find the rank of $y^2 = x^3 + 1918x$.

Let's do two more things related to ranks, and then start talking about some cool applications of elliptic curves.

Example: Use Magma to compute the image of α and $\bar{\alpha}$ for $C : y^2 = x^3 - 642x^2 + x$.

First, let's do α . The image of α must be contained in square classes corresponding to squarefree divisors of $b = 1$, so $\text{im } \alpha \subseteq \{\pm 1\}$. We have $\alpha(\mathcal{O}) = 1$ and $\alpha(T) = 1$ too. Is $-1 \in \text{im } \alpha$?

Here, we can look at the curve

$$N^2 = -M^4 - 642M^2e^2 - e^4.$$

The right hand side is < 0 and so the left hand side is ≥ 0 and so there are no non-trivial solutions. So $\text{im } \alpha = \{1\}$.

We have $\bar{a} = -2a = 1284$ and $\bar{b} = a^2 - 4b = 412160$. I will note that $4 \mid \bar{a}$ and $16 \mid \bar{b}$ and so we can simplify the equation for $\bar{C}$ by dividing $\bar{a}$ by 4 and dividing $\bar{b}$ by 16. This gives

$$\bar{C} : y^2 = x^3 + 321x^2 + 25760x.$$

The factorization of 25760 is $2^5 \cdot 5 \cdot 7 \cdot 23$, so

$$\text{im } \bar{\alpha} \subseteq \{\pm 1, \pm 2, \pm 5, \pm 7, \pm 23, \pm 10, \pm 14, \pm 46, \pm 35, \pm 115, \pm 161, \pm 70, \pm 230, \pm 322, \pm 805, \pm 1610\} = S.$$

We have that $\bar{\alpha}(\mathcal{O}) = 1$ and $\bar{\alpha}(\bar{T}) = 25760 \equiv 1610 \pmod{(\mathbb{Q}^\times)^2}$. Notice that $b = 1$, and since $16b = \bar{b} = \bar{a}^2 - 4\bar{b}$ is a square, the discriminant of $x^2 + 321x + 25760$ is a square (in fact it is 1). This means that this quadratic factors, and it factors as $(x + 160)(x + 161)$. So we have the points of order 2 $(-161, 0)$ and $(-160, 0)$ on $\bar{C}$. We have $\bar{\alpha}((-161, 0)) \equiv -161$ and $\bar{\alpha}((-160, 0)) \equiv -160 \equiv -10 \pmod{(\mathbb{Q}^\times)^2}$. So the image of $\bar{\alpha}$ contains $\{1, -10, -161, 1610\}$.

What happens if we test $d = 2$? We want to find M , N and e so that

$$N^2 = 2M^4 + 321M^2e^2 + 12880e^4.$$

If we create the hyperelliptic curve

$$C_2 : y^2 = 2x^4 + 321x^2 + 12880$$

in Magma and search for points via

```
R<x> := PolynomialRing(Rationals());
f := 2*x^4 + 321*x^2 + 12880;
C2 := HyperellipticCurve(f);
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`Points(C2 : Bound := 1000);`

Magma finds the points $(M, N, e) = (\pm 8, \pm 204, 1)$. So 2 is in the image of $\bar{\alpha}$ (and by closure, products of 2 and the four elements we already know are in the image of $\bar{\alpha}$).

This gives us that $\{1, 2, -5, -10, -161, -322, 805, 1610\}$ are contained in the image of α .

What happens if we test $d = 7$? We want to find M, N and e so that

$$N^2 = 7M^4 + 321M^2e^2 + 3680e^4.$$

If we create the hyperelliptic curve

$$C_7 : y^2 = 7x^4 + 321x^2 + 3680$$

in Magma and search for points, we don't find any.

We can test to see if C_7 has points mod every power of every prime with the command

`IsLocallySolvable(GenusOneModel(C7));`

This returns true - we cannot use modular arithmetic to rule out the existence of points on C_7 !

We could search for points with a higher bound

`Points(C7 : Bound := 1000000);`

it takes my computer 34 seconds to run the search, and it does not find any points.

So is 7 in the image of $\bar{\alpha}$?

Well, Magma can tell us that the rank of $y^2 = x^3 - 642x^2 + x$ is 1. (And also, the second return value of the rank command is true, which tells us Magma is sure it found the rank.)

This means that

$$2 = 2^r = \frac{|\text{im } \alpha| |\text{im } \bar{\alpha}|}{4}.$$

And so $|\text{im } \alpha| |\text{im } \bar{\alpha}| = 8$. Since we know $|\text{im } \alpha| = 1$, we must have that $|\text{im } \bar{\alpha}| = 8$. Thus, there are no rational points on

$$y^2 = 7x^4 + 321x^2 + 3680.$$

(Later this semester we'll get a bit more into how Magma can compute the rank in this situation.)

Cryptography

Section 1 - Intro to cryptography

Cryptography is the art and science of transmitting information in such a way that only the receiver can know it. The subject has a long history, and if you're interested in learning about it, I recommend taking MTH 243/682 if you haven't already. (It should be offered in Fall 2026.)

In the modern age, people often need to communicate securely even if they haven't met or shared anything in advance. This happens most often on the internet when files need to be transferred, or when it is necessary to conduct financial transactions online. Ideally, it is nice if this can be done quickly.

There are two types of cryptosystems:

i) Public-key cryptosystems - These are cryptosystems that do not require any previous communication between the two parties. These cryptosystems involve invertible functions that are easy to compute in one way, but extremely difficult (but theoretically possible) to compute in the opposite way. The security relies on the difficulty of computing them in the other way. These systems are typically somewhat slow, and so they are often used only to communicate a key for a *private-key* cryptosystem.

Examples: RSA, El Gamal cryptosystem, Diffie-Hellman (not quite a cryptosystem, but a key exchange protocol).

ii) Private-key cryptosystems - These are cryptosystems that rely on a shared key K . They only function if the two people communicating already know the key. These are typically much faster than public-key algorithms, and are used in practice. I'm not going to say much about these. Here are some examples:

DES - The data encryption standard. Approved as a standard in 1976. Used extensively from 1976 until the late 1990s. One instance broken in 1997. In 1998, custom-built hardware could break a DES key in 56 hours. In 2002 (!), it was replaced.

RC4 - Simple and speedy. Proposed by Ron Rivest in 1987. (RC means "Ron's Code" or "Rivest Cipher"). Previously used in WEP, WPA, SSL and TLS. Beginning in 2001, a series of stronger and stronger attacks against RC4 were found. It was prohibited for use in TLS in 2015, and disabled in most web browsers in early 2016. However, Microsoft Active Directory still allows its use and the vulnerability led to a ransomware attack in 2024.

AES - Very popular for private-key cryptography. Invented around 2000 when it became clear that DES wasn't secure anymore. There are three versions with 128 bit, 192 bit and 256 bit keys. NSA authorized the 256 bit version for transmission of top secret information. The best theoretical attacks are better than brute force by a factor of 4. Brute force would require billions of years (for a 128 bit key).

AES is vulnerable to side-channel attacks. (Measuring the time and power used by a computer when performing AES.) This requires access to computers performing the algorithm, though.

Section 2 - RSA and Diffie-Hellman

RSA

Alice wants to send a message to Bob.

Bob picks a large number N (say with 620 digits, or 2048 bits) that is a product of two large primes $N = pq$. (Why not more primes?) Bob chooses a number e (not too small) with $\gcd(e, (p-1)(q-1)) = 1$. (In most applications, $e = 65537$ is used.) Bob computes x and y so that $ex + (p-1)(q-1)y = 1$ and let $d = x \bmod (p-1)(q-1)$.

Bob shares the numbers N and e . Bob keeps p , q and d secret.

Alice takes her message and represents it as a number $M < N$. Alice computes $M^e \bmod N$ and sends it to Bob.

Bob takes the result $X \equiv M^e \bmod N$ and computes $X^d \bmod N$. This equals M .

Why does it work?

The set $(\mathbb{Z}/N\mathbb{Z})^\times$ is the multiplicative group modulo N . Its elements are $\{a \in \mathbb{Z} : 1 \leq a \leq N, \gcd(a, N) = 1\}$. The size of this group is called $\phi(N)$. For a general positive integer N ,

$$\phi(N) = N \prod_{p|N} (1 - 1/p),$$

and so in this case, $\phi(N) = pq(1 - 1/p)(1 - 1/q) = (p-1)(q-1)$.

Lagrange's theorem says that if G is a finite group, and $g \in G$, then the order of g divides the order of G . Thus, $g^{|G|} = 1$, the identity in G . In the context of $(\mathbb{Z}/N\mathbb{Z})^\times$, we get Euler's theorem.

Euler's theorem states that if $\gcd(a, n) = 1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$.

RSA works because $M^{de} = M^{1-(p-1)(q-1)y} \equiv M^1 \cdot M^{(p-1)(q-1)(-y)} \equiv (M^{(p-1)(q-1)})^{-y} \equiv M \cdot 1 \equiv M \pmod{N}$.

(You might be worried that if $\gcd(M, N) \neq 1$, then things break. It turns out that you can show that $M^{de} \equiv M \pmod{N}$ is still true even if $\gcd(M, N) > 1$.)

RSA is secure because finding d is essentially equivalent to finding p and q . Factoring large numbers turns out to be surprisingly hard. (More on this later) Normally, when RSA is used, the message is a key for AES (that has been padded with random bits to make it 2048 bits long or so).

Classical Diffie-Hellman

This is not a message transfer mechanism, it is only a key exchange procedure. It is very similar in flavor to the elliptic curve Diffie-Hellman algorithm (There is a related message transfer system called the El Gamal public-key cryptosystem.)

1. Alice and Bob agree on a prime number p and a number $\alpha \bmod p$ so that the order of α in $(\mathbb{Z}/p\mathbb{Z})^\times$ is large. (So α and p are public.)
2. Alice chooses a random x with $1 \leq x \leq p-2$ and Bob chooses a random y with $1 \leq y \leq p-2$. (Alice keeps x secret and Bob keeps y secret.)

3. Alice computes $\alpha^x \bmod p$ and sends it to Bob. (So α^x is public.)
4. Bob sends $\alpha^y \bmod p$ to Alice. (So α^y is public.)
5. Alice computes $(\alpha^y)^x \equiv \alpha^{xy} \pmod{p}$. Bob compute $(\alpha^x)^y \equiv \alpha^{xy} \pmod{p}$.

This number $\alpha^{xy} \pmod{p}$ is the shared key.

The security of this problem is based on solving the discrete logarithm problem. That is, given $\alpha \bmod p$ and $\alpha^x \bmod p$, how does one find x ?

Section 3 - Elliptic curve algorithms

The elliptic curve Diffie-Hellman algorithm works in a very similar way. Instead, a prime p is picked, and an elliptic curve $E/\mathbb{F}_p$ is used. The set $E(\mathbb{F}_p)$ is a group, and this group is used instead of $(\mathbb{Z}/p\mathbb{Z})^\times$.

A prime p , a curve E and a point $P \in E(\mathbb{F}_p)$ is agreed in advance. These are shared.

Alice chooses a random number α of size around p .

Bob chooses a random number β of size around p .

Alice computes αP and sends it to Bob. Bob computes βP and sends it to Alice.

Knowing βP and α , Alice computes $\alpha\beta P$ by adding βP to itself α times in the group law.

Knowing αP and β , Bob computes $\alpha\beta P$ by adding αP to itself β times in the group law.

The x -coordinate of $\alpha\beta P$ is the shared key.

The National Institute of Standards and Technology selected several particular elliptic curves to be used for cryptography. In 2013, concerns were raised that there might be a backdoor in NIST's P-256 curve, and Daniel Bernstein's Curve 25519 was used. The curve is

$$y^2 = x^3 + 486662x^2 + x$$

and the point P is chosen to have an x -coordinate of 9. The prime p used is $p = 2^{255} - 19$. This is used to transmit share a 128 bit AES key. (This curve was later "adopted" by NIST).